SENTINEL: AN INTEGRATED SECURITY OPERATIONS MANAGEMENT SYSTEM FOR DAVAO SECURITY & INVESTIGATION AGENCY, INC.

A Capstone Project
Proposal Presented to the Faculty of the
Information and Communications Technology
Program STI College Tagum

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Systems

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **SENTINEL: An Integrated Security Operations Management System**

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APPROVAL SHEET

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INTRODUCTION

Project Context

The private security global industry is transitioning toward automated, high-integrity digital ecosystems to address the discovery lag inherent in human-led monitoring. Global research in workforce capacity indicates that manual, reactive responses to personnel gaps significantly increase vacancy time, which remains a primary driver of security breaches in critical infrastructure (Shiyanbola et al., 2023). This movement is supported by modern performance management theories which emphasize that transparent, data-driven tracking is essential to maintaining accountability and preventing social loafing, a phenomenon where individual reliability decreases in the absence of structured, real-time monitoring (Aguinis, 2022).

The necessity for these technological advancements is echoed in recent Philippine legislation through the Private Security Services Industry Act, also known as Republic Act No. 11917 (2022). This law introduced a regime of professionalized accountability where agencies face strict liability for administrative negligence. Under the law's Implementing Rules and Regulations, agencies can be fined up to P5,000,000 or face license revocation for deploying personnel with expired licenses or unauthorized firearms (Jur.ph, 2025). Furthermore, national studies on the effectiveness of Philippine security guards identify that absenteeism and the abandonment of posts are most prevalent in agencies that lack advanced patrol monitoring and digital reporting systems (Abad, 2025).

In the specific context of the Davao Region, the Davao Security and Investigation Agency, Inc. operates in a landscape that has prioritized a culture of security to sustain its position as a leading safe haven in Southeast Asia (PIA, 2025). According to the Davao Regional Development Plan 2023 to 2028, the region's rapid economic growth has increased the demand for advanced technology and innovation to ensure peace and security in high-traffic logistics and commercial hubs like Tagum City (NEDA XI, 2024). Recent local research conducted in the Davao Region specifically highlights that compensation, punctuality, and alertness are the primary determinants of guard performance, yet many regional agencies still struggle with inattentiveness and abandonment of posts due to inconsistent manual oversight (Ondos and Origines, 2025). For DASIA Tagum, this represents a critical vulnerability where site vacancies may go undetected for hours, directly compromising the safety of local clients.

SENTINEL is an integrated digital ecosystem designed to resolve these cumulative vulnerabilities by centralizing personnel profiles, firearm telemetry, and shift logs into a high-performance single source of truth. The system works by embedding real-time compliance checks into the daily operational workflow. Before a guard can be deployed, the system automatically verifies their license validity and firearm authorization against central records. To manage high-concurrency requests from hundreds of guards, SENTINEL utilizes the Rust programming language and the Axum framework, providing native memory safety and asynchronous performance that prevents the data races typical of legacy systems (Scofield, 2025). When an operational gap such as a missed check-in is detected, the system immediately flags the vacancy and identifies the nearest qualified replacement, ensuring that the agency moves from a reactive stance to a proactive, legally defensible model of security management.

Purpose and Description

The primary purpose of this capstone project is to deliver a fully functional, production-ready Integrated Security Operations Management Platform that consolidates guard personnel management, equipment allocation, vehicle operations, and access control into a unified system for Davao Security & Investigation Agency, Inc.

SENTINEL is a web-based system that integrates operational data and workflows using a Rust/Axum backend API, a PostgreSQL relational database, and a React/TypeScript frontend deployed through Docker. The system is designed as an integrated operations platform that aligns with ERP and WFM principles by centralizing data, enforcing consistent workflows, and providing real-time operational visibility.

The platform implements four core modules. Guard Management covers personnel profiles, scheduling, attendance tracking, performance analytics, and replacement coordination to ensure uninterrupted site coverage. Equipment Management maintains firearm inventory, allocation workflows, permit records, and maintenance history to support regulatory compliance and accountability. Vehicle Operations manages armored vehicle assets, driver assignments, and trip tracking to improve deployment oversight. Access Control enforces role-based permissions, authentication, and audit logging to ensure that only authorized roles can access sensitive operations and data. The system has been validated through a 24-day operational simulation covering 24 distinct business scenarios and resolving 15 or more production issues.

Objectives

General Objectives

Deliver a fully implemented, tested, and production-ready Integrated Security Operations Management Platform that unifies personnel management, equipment allocation, vehicle operations, and access control into a single robust system with proven reliability through comprehensive operational simulation.

Specific Objectives

1. Implement a complete account and identity lifecycle module for all users.
    a. Provide user registration, email verification, resend verification, and secure login.
    b. Implement forgot password, reset code verification, and password reset workflows.
    c. Support profile management, profile photo update/delete, and role-based user creation.
2. Implement centralized personnel administration and approval workflows.
    a. Manage users by role: superadmin, administrator, supervisor, and guard.
    b. Provide pending guard approval queues and approval status updates.
    c. Support user update, deactivation/delete, role/status filtering, and guard-specific listings.
3. Implement scheduling, attendance, and workforce continuity automation.
    a. Create, update, delete, and list shifts with conflict-aware assignment.
    b. Enable guard check-in/check-out with attendance history tracking.
    c. Detect no-shows, request replacements, accept replacements, and manage guard availability.

4. Implement firearm inventory, issuance, and custody control.
    a. Provide full firearm CRUD with serial/model/caliber/status tracking.
    b. Implement firearm issuance, return, active allocations, and overdue allocation monitoring.
    c. Maintain firearm maintenance schedules, pending maintenance, and completion records.
5. Implement permit and training compliance management.
    a. Create and retrieve guard permits, including expiring and revoked permits.
    b. Support auto-expire permit processing for compliance enforcement.
    c. Manage training records, guard training history, and expiring training alerts.
6. Implement armored vehicle fleet and trip operations.
    a. Provide armored car CRUD, issuance/return, and active allocation monitoring.
    b. Manage driver assignment/unassignment and driver-vehicle linkage.
    c. Support trip creation, trip status updates, trip completion, and trip detail/history views.
    d. Manage vehicle maintenance scheduling, completion, and maintenance records.
7. Implement mission operations and merit-based performance evaluation.
    a. Enable mission assignment and mission retrieval for operational deployment.
    b. Calculate guard merit scores and provide ranked guard outputs.
    c. Support client evaluation submission, guard evaluation retrieval, and overtime candidate identification.
8. Implement support, communication, and notification services.
    a. Enable support ticket creation and guard ticket retrieval workflows.
    b. Provide user notifications, unread counts, mark-read/mark-all-read, and delete actions.

   c. Support alerting for operational events and decision-critical updates.

9. Implement incident management and real-time tracking intelligence.
   a. Provide incident creation, active incident retrieval, and incident status updates.
   b. Implement guard heartbeat, tracking point capture, and map data endpoints.
   c. Provide client site CRUD and shift proximity alert checks.
   d. Enable real-time websocket tracking stream for live operational visibility.

10. Implement analytics, predictive intelligence, and decision support.
   a. Provide analytics overview, trends, and guard reliability endpoints.
   b. Implement predictive alerts for permit expiry, maintenance risk, no-show patterns, and guard capacity risk.
   c. Provide AI (deterministic) services for guard absence risk, replacement suggestions, vehicle maintenance risk, incident classification, and incident summarization.

11. Implement real-time location monitoring and map-based operations.
   a. Provide OpenStreetMap-based visualization of guards, routes, and client sites.
   b. Implement live tracking updates through websocket streaming.
   c. Support map data retrieval and proximity-based operational alerts.

12. Implement governance, security, and auditability mechanisms.
   a. Maintain write-request audit logs for critical actions.
   b. Support session/token-based authenticated API access and presence/last-seen tracking.
   c. Provide health-check endpoints for service monitoring and operational readiness.

Scope and Limitations

Scope

The SENTINEL system is a web-based platform intended to strengthen operational control and compliance for a private security agency. The system is scoped to security agency operations and the regulatory environment set by the Private Security Services Industry Act (Republic Act No. 11917, 2022).

Data

- Personnel profiles, role/approval data, licensing and permit records, training records, schedules, attendance logs, availability status, and performance/merit metrics
- Firearm inventory, firearm allocations, firearm maintenance records, and firearm permit compliance data
- Armored vehicle assets, car allocations, driver assignments, trip records, and vehicle maintenance history
- Missions, incidents, support tickets, notifications, predictive alerts, tracking/map points, and audit/access logs

Process

- Account lifecycle workflows including registration, email verification, authentication, password reset, and profile management
- Shift scheduling, attendance validation, no-show detection, replacement request/acceptance, and supervisor oversight
- Firearm issuance/return workflows with permit validation, maintenance scheduling, and custody traceability
- Vehicle allocation, driver assignment, trip lifecycle management, and

preventive/corrective maintenance tracking

- Incident reporting, support ticket handling, notification delivery, and real-time operations tracking

People

- Superadmin, Administrator, Supervisor, and Guard roles with distinct permissions and dashboard views

Technology

- Web-based application accessible on desktop and mobile browsers
- React + TypeScript frontend and Rust + Axum backend services
- Centralized PostgreSQL database with relational integrity, auditability, and role-based data control
- Dockerized deployment and API-driven architecture with real-time websocket support for tracking views

Limitation of the Study

- The system does not integrate with external payroll, HR, or government licensing systems.
- Dedicated hardware-grade vehicle telematics integration is not yet implemented; current tracking relies on application-provided location updates and available device/network conditions.
- AI/predictive outputs are decision-support recommendations and do not autonomously execute final operational actions.
- Direct integration with third-party hardware ecosystems (CCTV, IoT sensors, access control turnstiles) is not included.
- Offline-first field operation is limited; core workflows depend on network/API availability.

Review of Related Literature/Studies/Systems

Related Literature

The Philippine private security sector is currently navigating its most significant legal transition in over fifty years. The enactment of the Private Security Services Industry Act (Republic Act No. 11917, 2022) effectively repealed the outdated RA 5487, shifting the industry toward a regime of "Professionalized Accountability." According to Jur.ph (2025), the law's Implementing Rules and Regulations (IRR) mandate that Private Security Agencies (PSAs) maintain highly accurate, digitized records of License to Exercise Security Profession (LESP) and firearm permits. The act introduces a "strict liability" framework where administrative negligence, such as deploying an unlicensed guard, can result in fines ranging from ₱50,000 to ₱100,000 per violation, or up to ₱5,000,000 for agency-wide license failures.

This regulatory pressure has necessitated a move toward Automated Regulatory Compliance Tracking (ARCT). Research by Khinvasara, T., Shankar, A., & Wong, C. (2024) suggests that for organizations managing complex, shifting rules, the use of automated monitoring is a "significant advancement" that eliminates the "discovery lag" inherent in manual audits. By leveraging system-driven triggers for license expirations, PSAs can transition from reactive compliance to a proactive stance, ensuring that no personnel or asset is deployed without valid legal authorization.

Operational continuity in the private security sector is heavily dependent on reliable guard presence. As identified in the operational risks of DASIA Tagum, "No-Show" incidents represent a critical vulnerability where manual processes relying on phone calls and delayed human intervention fail to address "discovery lag." Research in workforce capacity

optimization emphasizes that manual, reactive responses to workforce gaps significantly increase "vacancy time," directly correlating with higher security breach probabilities (Shiyanbola et al., 2023).

Modern systems mitigate this by automating the identification and deployment of replacements based on real-time availability, shifting the organizational stance from reactive crisis management to proactive risk mitigation (Al-Khafajiy et al., 2022). Furthermore, advanced algorithms now enable "predictive scheduling," which analyzes historical No-Show data to alert administrators of potential gaps before they occur, drastically reducing the operational vacancy time (TrackTik, 2025).

The transition from inconsistent, manual tracking to a structured, data-driven framework is essential for maintaining Distributive Justice, the perceived fairness of how shifts, rewards, and responsibilities are allocated within an organization. Aguinis (2022) posits that the absence of a structured performance management framework leads to "Social Loafing," a psychological phenomenon where employees decrease effort due to a perceived lack of individual accountability.

For security agencies like DASIA Tagum, implementing centralized performance metrics such as punctuality and client feedback logs ensures that shift allocation is based on objective data rather than subjective selection. This transparency not only increases overall guard reliability but also fosters a culture of accountability, as guards understand that their performance is directly linked to future opportunities (Tan et al., 2024).

The systematic allocation of firearms is a matter of both Chain of Custody (CoC) and legal necessity under the Private Security Services Industry Act (Republic Act No. 11917, 2022). The law mandates strict accountability for firearms, requiring that only authorized, trained, and compliant guards carry weapons. Manual processes often suffer from "data drift," where the status of a weapon and the permit validity of the assigned guard are not reconciled in real-time, leading to potential administrative negligence. Centralized digital tracking is identified as the primary defense against the misallocation of high-risk equipment and is a prerequisite for generating the legally defensible audit logs required by the National Cybersecurity Plan 2022-2028 (DICT, 2024; PNP-SOSIA, 2022).

The selection of the backend technical stack is a core security decision for mission-critical systems. Scofield, M. B. (2025) argues that Rust's "ownership and borrowing" model provides a fundamental architectural advantage: Memory Safety without a Garbage Collector. This ensures that SENTINEL is natively resistant to buffer overflows and data races, vulnerabilities that often plague systems handling high-concurrency data like real-time firearm tracking.

Complementing the language is the Axum framework, which is built on the Tokio asynchronous runtime. As highlighted by NashTech (2025), Axum is designed for high-performance web services where scalability and raw speed are non-negotiable. For SENTINEL, this means the Attendance Tracking and No-Show Detection modules can handle hundreds of simultaneous "check-in" requests during shift rotations without performance degradation. This "asynchronous-first" design allows the system to remain responsive even when processing complex relational logic across personnel, vehicle, and firearm databases.

In an Integrated Operations Platform, the database must serve as the immutable "Single Source of Truth." PostgreSQL (2025) is recognized as the premier open-source RDBMS for enterprise planning due to its strict adherence to ACID (Atomicity, Consistency, Isolation, Durability) properties. Nguyen, R. (2025) emphasizes that PostgreSQL's implementation of Foreign Key Constraints and Unique Exclusion Constraints is the primary defense against "data drift."

In the SENTINEL ecosystem, these constraints ensure that a firearm or armored vehicle cannot be logically "double-booked" or assigned to a guard who does not exist in the personnel table. This high level of relational integrity is a prerequisite for generating the legally defensible Audit Logs required by the National Cybersecurity Plan 2023-2028 (DICT, 2024), ensuring that every asset movement is tied to a verifiable digital identity.

Related Studies and/or Systems

Empirical research confirms that automated oversight is a primary driver of operational integrity. A study by Atlam, H. F., & Yang, Y. (2025) found that organizations utilizing unified Access Control and Resource Planning (ERP) systems saw a 27% drop in unauthorized access violations and reclaimed 36% of staff time previously lost to manual verification. These findings suggest that hardwiring compliance into the system architecture, rather than treating it as an afterthought, tightens process integrity and reduces the risk of internal data tampering.

Furthermore, research by Shiyanbola, J. O., et al. (2023) on "Workforce Capacity Optimization" highlights a flaw in traditional security: "discovery lag." This occurs when supervisors only identify a personnel gap after a shift has failed. Their study demonstrates that real-time, asynchronous detection models allow for a proactive 11.8% improvement in site coverage reliability. Locally, ResearchGate (2025) cites a study by Respicio (2023) which found that Philippine security guards operating under digital monitoring exhibited significantly higher levels of alertness and punctuality because the system provided a transparent, inescapable layer of accountability that manual logs could not match. The functional blueprint of SENTINEL is informed by the performance of several industry-leading platforms such as TrackTik and Guardhouse.

TrackTik

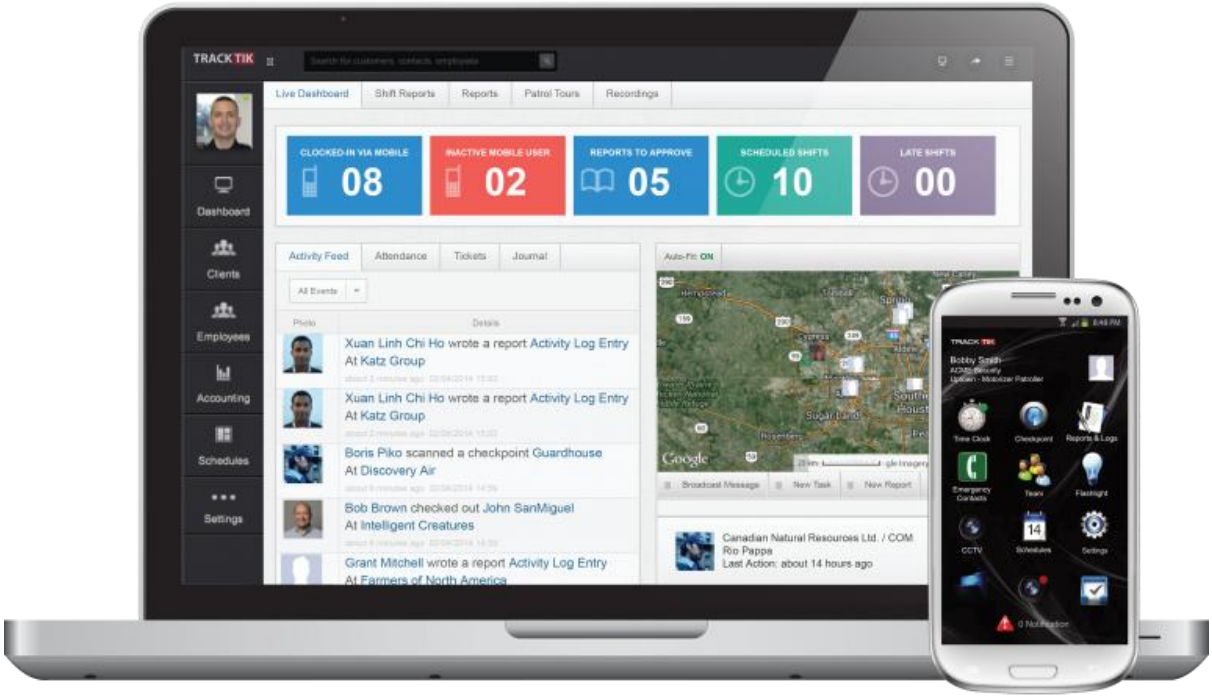

Figure 1. TrackTik

TrackTik (2025) has redefined security workforce management by moving beyond static scheduling toward an integrated, AI-driven operational model. According to the Trackforce 2025 Physical Security Operations Benchmark Report, the industry is currently facing an "AI adoption paradox" where the value of predictive scheduling is recognized, but cost remains a barrier for mid-sized firms. TrackTik's success is anchored in its ability to consolidate alarm monitoring, guard dispatch, and business administration into a single 24/7 resilient architecture, which users report has increased operational efficiency by approximately 20%. This efficiency is primarily attributed to the reduction of manual administrative burdens through automated compliance tracking and real-time intelligence feeds (TrackTik, 2025).

Guardhouse

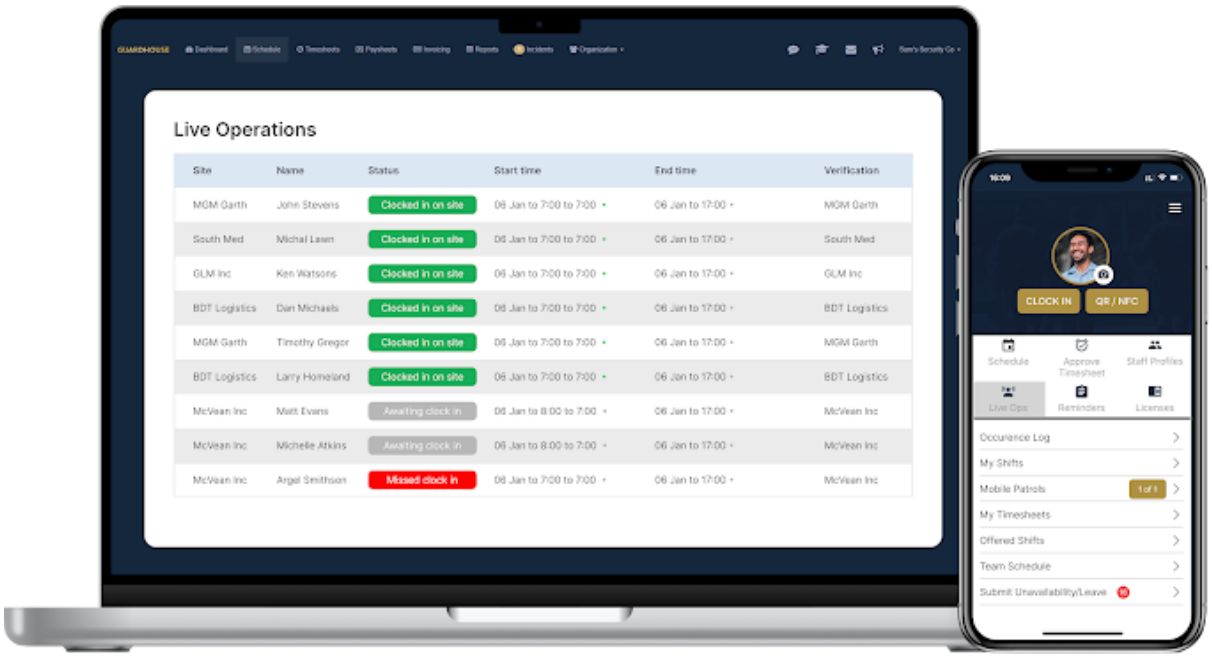

Figure 2. Guardhouse

Guardhouse (2026) addresses the specific administrative "overload" that has become a measurable barrier to performance in the mid-2020s. Research from NODE Magazine (2026) indicates that security employees often lose an average of 15 hours per week to routine administrative tasks, such as re-entering data across fragmented systems. Guardhouse mitigates this by unifying scheduling, GPS tracking, and invoicing into a streamlined workflow, reportedly reducing office-based administrative time by 40–60%. Its "Confidence in Compliance" module provides a specialized framework for daily license verification, a logic that SENTINEL adapts to specifically meet the stringent requirements of RA 11917 (GetApp, 2026).

MySecuritas

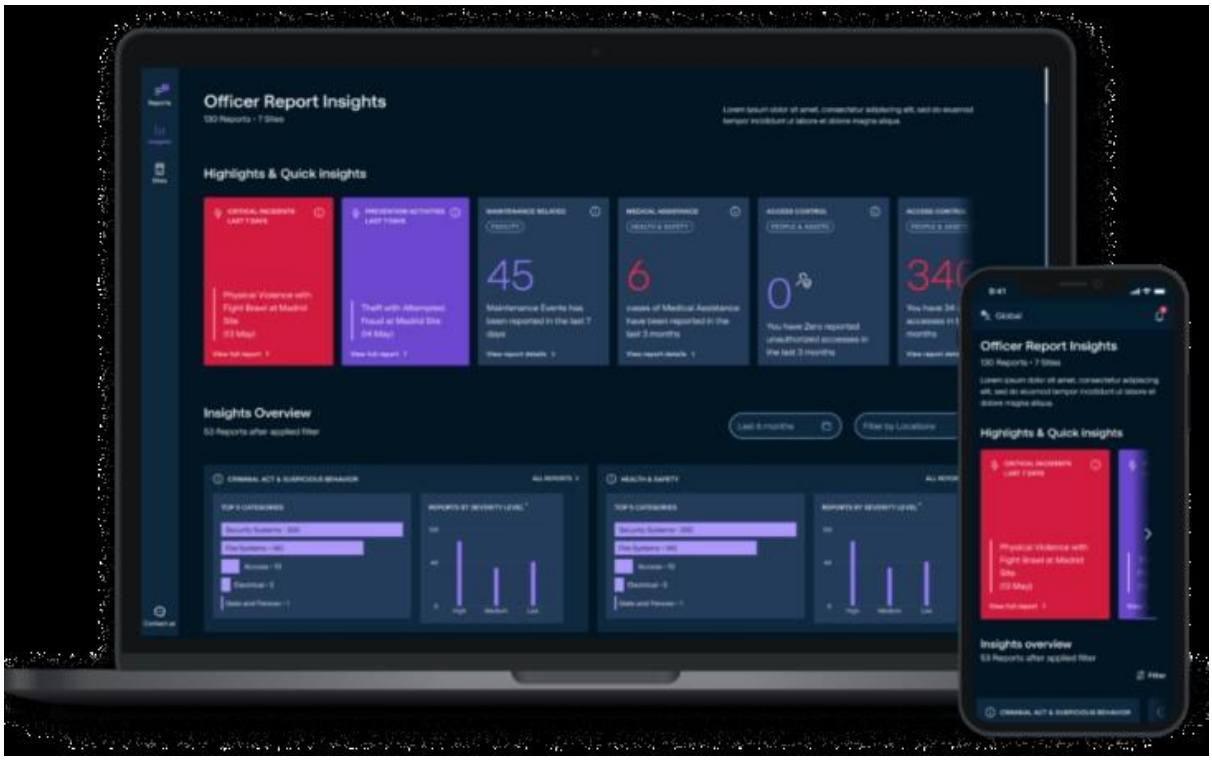

Figure 3. MySecuritas

Securitas "MySecuritas" (2026) represents the industry's shift toward "Situational Understanding" and proactive risk management. The Securitas 2026 Global Technology Outlook Report identifies that the rapid advancement of generative AI for contextual understanding is now a top priority for 30% of security decision-makers (Securitas Technology, 2025). MySecuritas operationalizes this by providing a unified dashboard that transforms disparate incident reports into actionable statistics and trend highlights. This allows supervisors to identify risk patterns before they escalate, a core design principle mirrored in SENTINEL's Performance Analytics Dashboard (Securitas, 2026).

Silvertrac Software

Figure 4. Silvertrac Software

Silvertrac Software (2026) is widely utilized for "Proof of Performance" and on-site accountability, particularly in parking and property management environments. While it excels in field reporting and checkpoint verification via mobile applications, its architectural focus remains on incident management and officer accountability (Silvertrac Software, 2026). Technical evaluations of generic guard management systems suggest a persistent "visibility gap" in specialized asset tracking; most platforms lack the deep relational logic required for a Digital Chain of Custody for high-risk assets like firearms and armored car fleets (Hardcat, 2025). SENTINEL fills this niche by integrating these high-stakes logistics, traditionally managed in siloed armory or fleet software, into the primary workforce management ecosystem.

Across the reviewed literature, several consistent themes emerge. Researchers emphasize the importance of integrating operational data to improve organizational visibility and decision-making. Studies from both Philippine and international contexts highlight the limitations of fragmented monitoring tools and manual processes.

Technological trends such as automated scheduling algorithms, attendance tracking, and operational dashboards are increasingly adopted across industries. However, existing systems often address only a single domain of operations, such as workforce scheduling or surveillance monitoring. Global systems like TrackTik and Guardhouse demonstrate that the industry standard is defined by Real-Time Accountability and Centralized Data Ownership. However, these systems often fail to accommodate the granular administrative nuances of Philippine law.

The reviewed literature therefore reveals a clear research gap in integrated systems that combine workforce management, compliance monitoring, and operational oversight within a unified platform tailored to security agencies. The SENTINEL system addresses this void by combining high-concurrency Rust/Axum architecture with modules specifically for Philippine-specific licensing (RA 11917), Firearm Custody, and Armored Car Fleet Management, allowing local agencies to achieve multinational levels of visibility and compliance. It provides a centralized solution that integrates scheduling automation, regulatory compliance verification, and real-time operational dashboards within a single enterprise system.

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